

DBMS – PRODUCT AND CATEGORY MANAGEMENT

Project Title: EcoCart – Sustainable Products E-Commerce Order Management System

1. Database Design

1.1 Product Table

Attribute	Key
Product_ID	Primary Key (PK)
Product_Name	-
Category_ID	Foreign Key (FK)
Price	-
Stock_Quantity	-

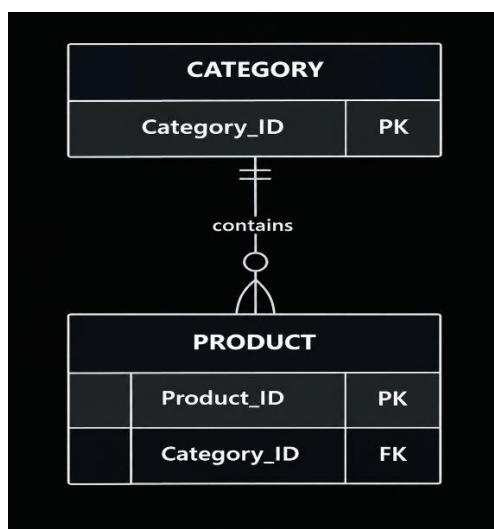
1.2 Category Table

Attribute	Key
Category_ID	Primary Key
Category_Name	-
Category_Description	-

2. Primary and Foreign Key Relationship

Category → Product

- Category.Category_ID → Primary Key
- Product.Product_ID → Primary Key
- Product.Category_ID → Foreign Key referencing Category.Category_ID
- Relationship: One-to-Many (1:M)



3. Table Creation

3.1 Create Category Table

```
SQL> CREATE TABLE Category (  
2     Category_ID INT PRIMARY KEY,  
3     Category_Name VARCHAR(100) NOT NULL UNIQUE,  
4     Description VARCHAR(200)  
5 );  
  
Table created.
```

3.2 Create Product Table

```
SQL> CREATE TABLE Product (  
2     Product_ID INT PRIMARY KEY,  
3     Product_Name VARCHAR(100) NOT NULL,  
4     Category_ID INT,  
5     Price NUMBER(10,2) NOT NULL,  
6     Stock_Quantity INT NOT NULL,  
7     FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID)  
8 );  
  
Table created.
```

4. Data Insertion

4.1 Category Records

```
SQL> INSERT INTO Category VALUES (1, 'Laptops', 'Portable computers');  
1 row created.  
  
SQL> INSERT INTO Category VALUES (2, 'Mobiles', 'Smartphones and mobile devices');  
1 row created.  
  
SQL> INSERT INTO Category VALUES (3, 'Accessories', 'Computer and mobile accessories');  
1 row created.  
  
SQL> INSERT INTO Category VALUES (4, 'Audio', 'Headphones and speakers');  
1 row created.
```

4.2 Product Records

```
SQL> INSERT INTO Product VALUES (101, 'HP Laptop', 1, 55000, 10);  
1 row created.  
  
SQL> INSERT INTO Product VALUES (102, 'Dell Laptop', 1, 62000, 8);  
1 row created.  
  
SQL> INSERT INTO Product VALUES (103, 'Samsung Galaxy', 2, 28000, 15);  
1 row created.  
  
SQL> INSERT INTO Product VALUES (104, 'Wireless Mouse', 3, 799, 30);  
1 row created.  
  
SQL> INSERT INTO Product VALUES (105, 'Bluetooth Speaker', 4, 1999, 20);  
1 row created.
```

5. Data Retrieval

5.1 Display Category Records

```
SQL> SELECT * FROM Category;
```

```
CATEGORY_ID
```

```
-----
```

```
CATEGORY_NAME
```

```
-----
```

```
DESCRIPTION
```

```
-----
```

```
1
```

```
Laptops
```

```
Portable computers
```

```
2
```

```
Mobiles
```

```
Smartphones and mobile devices
```

```
CATEGORY_ID
```

```
-----
```

```
CATEGORY_NAME
```

```
-----
```

```
DESCRIPTION
```

```
-----
```

```
3
```

```
Accessories
```

```
Computer and mobile accessories
```

```
4
```

```
Audio
```

```
CATEGORY_ID
```

```
-----
```

```
CATEGORY_NAME
```

```
-----
```

```
DESCRIPTION
```

```
-----
```

```
Headphones and speakers
```

5.2 Display Product Records

```
SQL> SELECT * FROM Product;
```

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID	PRICE	STOCK_QUANTITY
101	HP Laptop	1	55000	10
102	Dell Laptop	1	62000	8
103	Samsung Galaxy	2	28000	15
104	Wireless Mouse			
105	Bluetooth Speaker	3	799	30

7. 1 Product Updation

Update 1: Change HP Laptop price

```
SQL> UPDATE Product
  2 SET Price = 53000
  3 WHERE Product_Name = 'HP Laptop';

1 row updated.
```

Update 2: Change Samsung Galaxy stock

```
SQL> UPDATE Product
  2 SET Stock_Quantity = 20
  3 WHERE Product_Name = 'Samsung Galaxy';

1 row updated.
```

7.2 Product Deletion

Delete 1: Delete Wireless Mouse

```
SQL> DELETE FROM Product
  2  WHERE Product_Name = 'Wireless Mouse';

1 row deleted.
```

Delete 2: Delete Bluetooth Speaker

```
SQL> DELETE FROM Product
  2  WHERE Product_Name = 'Bluetooth Speaker';

1 row deleted.
```

Delete 3: Delete Dell Laptop

```
SQL> DELETE FROM Product
  2  WHERE Product_Name = 'Dell Laptop';

1 row deleted.
```

8. Category-wise Product Reports

8.1 Product Listing by Category

```
SQL> SELECT C.Category_Name, P.Product_Name, P.Price, P.Stock_Quantity
  2  FROM Category C
  3  JOIN Product P
  4  ON C.Category_ID = P.Category_ID
  5  ORDER BY C.Category_Name;
```

CATEGORY_NAME

PRODUCT_NAME

PRICE STOCK_QUANTITY

Laptops

HP Laptop
53000 10

Mobiles

Samsung Galaxy
28000 20

CATEGORY_NAME

PRODUCT_NAME

PRICE STOCK_QUANTITY

8.2 Products in Laptops Category

```
SQL> SELECT P.Product_Name, P.Price, P.Stock_Quantity
  2  FROM Product P
  3  JOIN Category C
  4  ON P.Category_ID = C.Category_ID
  5  WHERE C.Category_Name = 'Laptops';
```

PRODUCT_NAME

PRICE	STOCK_QUANTITY
HP Laptop	53000
	10

8.3 Product Count by Category

```
SQL> SELECT C.Category_Name, COUNT(P.Product_ID) AS Product_Count
  2  FROM Category C
  3  LEFT JOIN Product P
  4  ON C.Category_ID = P.Category_ID
  5  GROUP BY C.Category_Name;
```

CATEGORY_NAME

PRODUCT_COUNT

Laptops	1
Mobiles	1